

Mohamed Abdellatif Etman

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PROFESSIONAL SUMMARY

Innovative AI Engineer with expertise in designing and deploying end-to-end solutions using Generative AI, multimodal analysis, and cloud technologies. Proven ability to build and scale complex AI systems leveraging state-of-the-art models (GPT-4o, Gemini, Llama 3), RAG pipelines, and vector databases to solve real-world business problems. Adept at full-stack development, API design, and MLOps with strong focus on creating robust, scalable, and efficient applications on Microsoft Azure. Successfully delivered multiple AI-powered solutions and extensive portfolio of mobile applications driving measurable improvements in operational efficiency and user experience.

WORK EXPERIENCE

AI Engineer (Freelance)

December 2024 – Present

Independent Contractor

Remote

- Architected and delivered multiple end-to-end AI applications powered by Generative AI (GPT-4o, Gemini, Claude), designing high-precision RAG pipelines achieving exceptional accuracy in contextual data retrieval from large-scale document datasets.
- Engineered scalable multilingual NLP services using Azure Cognitive Services and custom transformer pipelines, processing high-volume monthly requests with enterprise-grade uptime and reliability.
- Developed asynchronous backend APIs using FastAPI with Redis session management, significantly reducing response latency and enabling concurrent handling of numerous user sessions with optimal performance.
- Created interactive monitoring dashboards using Streamlit with real-time analytics capabilities and containerized complete solutions using Docker for seamless deployment on Microsoft Azure infrastructure.
- Integrated WhatsApp Business API and Meta Graph API webhooks to deliver conversational AI experiences, achieving high user satisfaction scores across extensive customer interactions.

AI Developer – MCIT Innovation Project

November 2024 – March 2025

Ministry of Communications and Information Technology (MCIT)

Remote, Egypt

- Orchestrated a multimodal AI system for employee evaluation, synthesizing video, audio, and text analytics into actionable insights using Llama 3 and Qwen-VL, processing extensive evaluation sessions with high accuracy and reliability.
- Integrated OpenPose and MediaPipe for real-time behavioral analysis detecting multiple non-verbal cues and Whisper for multi-language audio transcription with excellent accuracy rates.
- Optimized inference pipelines for Azure Cloud deployment, substantially reducing processing time through advanced batch processing and model quantization techniques.
- Benchmarked Gemini 2.0 Flash against open-source alternatives (Llama 3.2, Qwen-VL) to maximize system efficiency, achieving significant cost reduction while maintaining high performance standards.
- Delivered startup prototype that received recognition from MCIT leadership, demonstrating innovation in government digital transformation initiatives.

GenAI Engineer (Internship)

July 2024 – October 2024

National Bank of Egypt (NBE)

Remote, Egypt

- Engineered the "Financial Dashboard Auto Transcript Analyzer," a production-ready tool utilizing Vision Transformers (Donut) to extract embedded data from numerous chart images with high accuracy.
- Implemented a hybrid question-answering engine combining GPT-NeoX-20B with FAISS vector retrieval to generate narrative financial summaries, substantially reducing analyst review time.
- Developed FastAPI microservice with Azure Text-to-Speech integration, enabling voice-enabled interactions for financial data queries supporting multiple concurrent users.
- Built desktop application using PyQt5 with intuitive GUI, deployed across multiple banking terminals for internal financial analysis workflows.

Full-Stack Developer & Mobile App Team Lead

August 2022 – Present

Camp Coding

Tanta, Egypt

- Led development team building cross-platform mobile applications using React Native, delivering extensive portfolio of apps with significant user adoption across iOS and Android platforms.
- Architected scalable backend services using PHP and MySQL, optimizing database queries and substantially improving load times for data-intensive applications.
- Managed complete software development lifecycle from requirements gathering to deployment, implementing Agile methodologies and sprint planning for efficient project delivery.
- Mentored junior developers on best practices in mobile development, API integration, and database design, significantly improving team code quality and development efficiency.

KEY PROJECTS

Generative AI System for Performance Analytics (MCIT Government Project)

- **Overview:** Co-engineered a comprehensive AI solution for the Ministry of Communications automating employee performance reviews by synthesizing multimodal data from video, audio, and text sources.
- **Key Contributions:** Orchestrated end-to-end pipelines combining computer vision (OpenPose, MediaPipe) for gesture recognition, audio analysis (Whisper) for speech transcription, and NLP for sentiment analysis. Optimized report generation by implementing Llama 3 and Qwen-VL to translate behavioral data into structured feedback, processing extensive evaluation sessions.
- **Impact:** Substantially reduced evaluation time, improved consistency in performance assessments, and provided data-driven insights for HR decision-making.
- **Technologies:** Python, PyTorch, OpenPose, MediaPipe, Whisper, Llama 3, Gemini 2.0 Flash, Azure Cloud, FastAPI, Docker.

RAG-Powered Conversational AI Assistant (Travel Agency Client)

- **Overview:** Designed and deployed complete backend architecture for an intelligent tourism chatbot capable of handling complex multi-turn conversations across WhatsApp and Web platforms, serving extensive customer interactions.
- **Key Contributions:** Architected high-precision RAG system using ChromaDB achieving excellent retrieval accuracy from large document collections. Engineered asynchronous API gateway using FastAPI to handle numerous concurrent requests with Redis session persistence. Integrated Meta Graph API webhooks for real-time WhatsApp message processing.
- **Impact:** Achieved high customer satisfaction scores, substantially reduced response time, and successfully handled significant portion of customer inquiries autonomously.
- **Technologies:** Gemini, GPT-4o, RAG, ChromaDB, LangChain, Azure Cognitive Services, FastAPI, Redis, Docker, WhatsApp Business API.

Automated Financial Insight Generator (National Bank of Egypt)

- **Overview:** Developed production-grade desktop application utilizing Vision Transformers to extract and analyze financial data from chart images, enabling natural language queries via voice and text interfaces.
- **Key Contributions:** Deployed Vision Transformers (Donut) achieving high accuracy in extracting numerical data from complex financial charts. Built intelligent Q&A system combining FAISS vector search with GPT-NeoX-20B for contextual financial analysis. Developed user-friendly PyQt5 desktop application with Azure TTS integration for voice-enabled experience.
- **Impact:** Substantially reduced financial analyst review time, processed extensive chart collections, and enabled faster decision-making for banking operations.
- **Technologies:** Transformers, Donut, FAISS, GPT-NeoX-20B, FastAPI, Azure Cognitive Services, PyQt5, Python.

CORE COMPETENCIES

Generative AI • Large Language Models • RAG Systems • MLOps • Azure Cloud • API Development • Computer Vision • Natural Language Processing • Full-Stack Development • FastAPI • Docker • Machine Learning • Deep Learning • Microservices • Vector Databases

EDUCATION

Diploma in AI, Data Science, and Cloud Computing January 2024 – December 2024
Digital Egypt Builders Initiative (DEBI), Ministry of Communication Hybrid, Egypt

- Intensive 12-month program covering Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, and MLOps.

Bachelor’s Degree in Computer Science July 2019 – July 2023
Faculty of Computers and Information, Tanta University Tanta, Egypt

- Relevant Coursework: Algorithms, Data Structures, Database Systems, Software Engineering, Artificial Intelligence, Machine Learning.

TECHNICAL SKILLS

Generative AI & LLMs: GPT-4o, Gemini, Claude, Llama 3, Qwen-VL, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, ChromaDB, Pinecone), LangChain, Prompt Engineering

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Transformers, BERT, CNNs, RNNs, LSTMs, Model Fine-tuning, Transfer Learning

Natural Language Processing: spaCy, NLTK, Hugging Face, Text Classification, Sentiment Analysis, Named Entity Recognition, Text Generation

Computer Vision & Multimodal AI: OpenCV, OpenPose, MediaPipe, YOLO, Vision Transformers, Image Classification, Object Detection

Audio Processing: Whisper, Praat, librosa, Speech-to-Text, Audio Feature Extraction, Voice Analysis

Programming Languages: Python (Advanced), JavaScript, TypeScript, PHP, SQL, Bash

Backend Development: FastAPI, Django, Flask, Node.js, RESTful APIs, WebSocket, Async Programming, Redis, Celery

ERP & Business Automation: Odoo API Integration (XML-RPC, JSON-RPC), Custom Module Development, ORM, Business Logic Implementation

Databases: MongoDB, MySQL, PostgreSQL, Vector Databases, Database Optimization, Query Performance Tuning

Frontend & Mobile Development: React Native, React.js, Streamlit, PyQt5, Responsive Design

Cloud Platforms & Services: Microsoft Azure (Cognitive Services, App Service, Functions, Blob Storage), Azure OpenAI, Azure ML

MLOps & DevOps: Docker, Docker Compose, Git, GitHub, CI/CD Pipelines, MLflow, Hugging Face Hub, Model Deployment, Monitoring

Tools & Technologies: Postman, Jupyter Notebook, VS Code, Linux, Agile Methodologies

CERTIFICATIONS

- **Microsoft Azure AI-102:** AI Engineer Associate – Certified in designing and implementing Azure AI solutions.
- **Stanford Machine Learning Specialization** (Coursera) – Andrew Ng’s comprehensive ML course covering supervised, unsupervised learning, and neural networks.
- **Deep Learning Specialization** (Coursera) – Advanced deep learning including CNNs, RNNs, LSTMs, and Transformers.
- **Generative AI with Large Language Models** (Coursera) – Specialized training in LLM architecture, fine-tuning, and deployment.
- **MLOps Tools & Best Practices** (DEBI) – Practical experience with MLflow, Hugging Face, Docker, and CI/CD for ML systems.

LANGUAGES

- **Arabic:** Native Speaker
- **English:** Professional Working Proficiency (Written and Spoken)

ADDITIONAL INFORMATION

- **Military Status:** Exempted
- **Availability:** Immediate
- **Work Authorization:** Egyptian Citizen, available for remote and on-site opportunities